

Towards an Intelligent Platform for Supporting Quality Assessment in Blockchain Systems



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Motivation & Context

Quality Challenge Software

Quality Challenge Software in blockchain systems involves:

- Maintainability, adaptability, and updatability, enabling software to evolve continuously (Zhu et al., 2022);
- Adjustment to constant changes in security, performance, and compliance requirements;
- Minimizing risks and promoting ecosystem reliability.

What makes blockchain different from conventional software (Porru et al., 2017):

- Immutability
- Decentralized execution
- Consensus protocols
- Transaction fees
- Domain-specific security threats

Most existing solutions focus primarily on **smart contracts vulnerabilities**, **benchmarking consensus mechanisms**, **throughput**, and **scalability** (Albshri et al., 2022).

IDENTIFIED GAPS

- Existing tools rarely address quality metrics as defined by **ISO/IEC 25010**;
- RAG-based assistants lack an **explicit quality model**; (Mansurova et al., 2023)
- Quality assessment also involves **social and collaborative aspects**, how practitioners apply metrics in practice.

Motivation & Context

Objective

This extended abstract presents the **methodological design** of a **support platform to assist** in the evaluation and application of **software quality metrics** in **blockchain-based** systems.

THE PLATFORM INTEGRATES TWO KEY COMPONENTS

- 1) A **metrics catalog** aligned with ISO/IEC 25010, gathering structured information on quality attributes, formulas, application criteria, and relevant elements;
- 2) A **chatbot to assist** users in identifying, organizing, and applying these metrics in blockchain contexts, powered by LLMs combined with **Retrieval-Augmented Generation (RAG)**.

Methodology

Design Science Research (DSR)

DSR focuses on the **creation and evaluation of artifacts** to address real-world problems in information technology (Vaishnavi and Kuechler, 2004).

- DSR is widely used to develop solutions applicable to both **academia and industry**

Vaishnavi, V. & Kuechler, W. (2004). Design research in information systems.

	PROCESS	METHOD
PHASE 1 PROBLEM AWARENESS	Identifying the state of the art of blockchain metrics	Systematic Mapping Study (SMS)
PHASE 2 SUGGESTION	Creationg of the catalog for blockchain metrics	Creation of the catalog based on the data collected through SMS
PHASE 3 DEVELOPMENT	Tool development with catalog metrics	Implementation of front-end, back-end, and integration with LLM models
PHASE 4 EVALUATION	Quantitative and Qualitative experimental evaluation	QUANTITATIVE EVALUATION performance analysis QUALITATIVE EVALUATION interviews & usability testing
PHASE 5 CONCLUSION	Reflection on general knowledge based on experimental evaluations	

Methodology - Design Science Research (DSR)

Phase 1: Problem Awareness - Systematic Mapping Study (SMS)

Following established guidelines (Petersen et al., 2008):

- **Sources:** Scopus, IEEE Xplore, ScienceDirect, ACM Digital Library
- **128 primary studies** analyzed
- **160 distinct metrics identified and classified (ISO/IEC 25010)**
- Published in **Saraiva et al. (2025)**.

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Petersen, K. et al. (2008). Systematic mapping studies in software engineering. EASE. Saraiva, R. et al. (2025). Metrics for Quality Assessment in Blockchain-based Systems: A SMS. SBSI.

Methodology - Design Science Research (DSR)

Phase 2: Suggestion - Metrics Catalog

Metrics catalog inspired by Olsina et al. (2002), based on SMS results (Saraiva et al., 2025) with each metric is **fully documented** with:

- **Definition** and **objective**
- **Formula** and **scale**
- **Application criteria** and usage protocols

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Olsina, L. et al. (2002). Designing a Catalogue for Metrics. ICWE, Springer.
Saraiva, R. et al. (2025). Metrics for Quality Assessment in Blockchain-based Systems: A SMS. SBSI.

Methodology - Design Science Research (DSR)

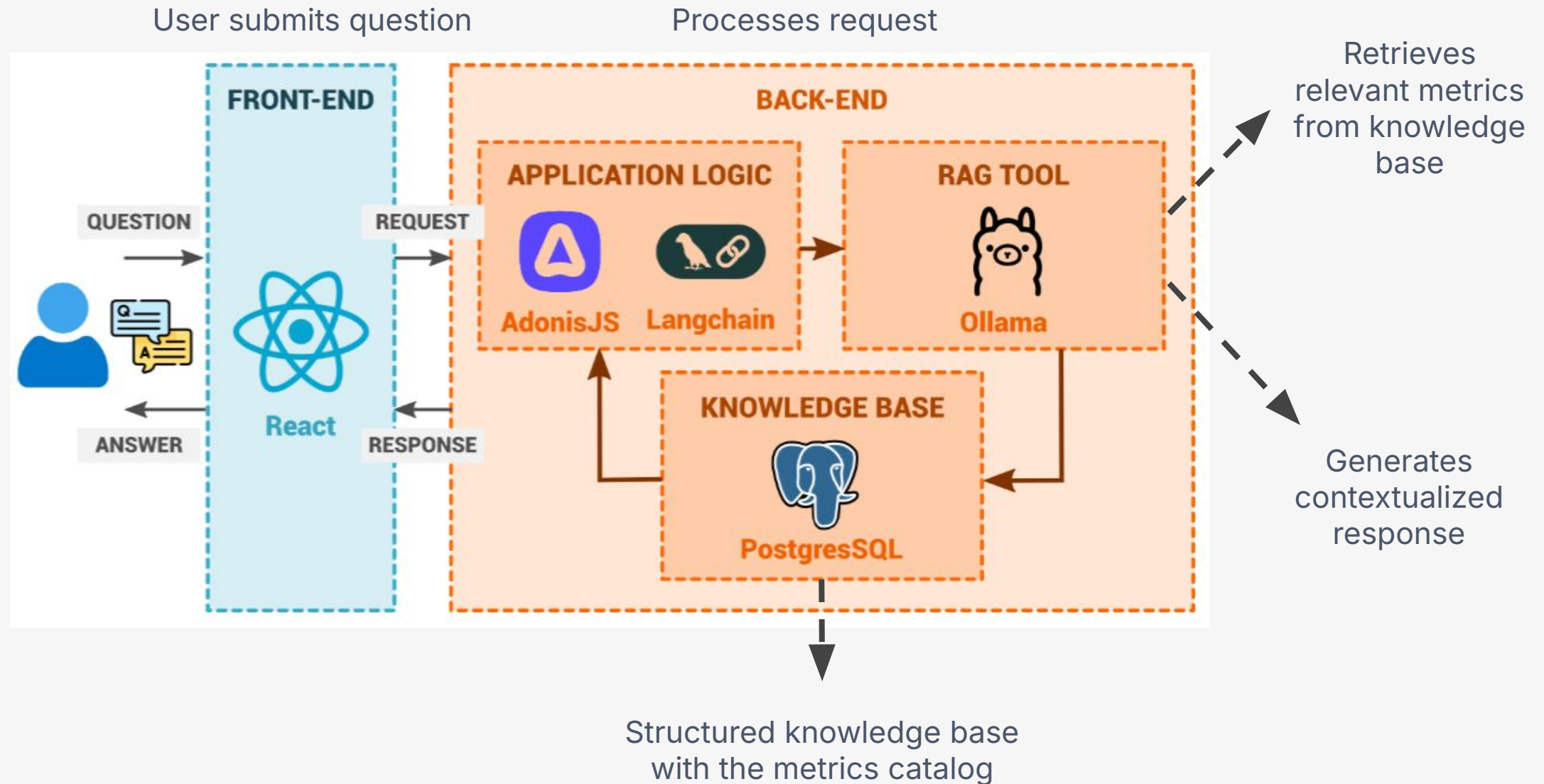
Phase 3: Development

The chatbot processes user queries by **retrieving relevant information** from the knowledge base via the **RAG component**, combining it with the language model to generate **contextualized responses** in real time.

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Methodology - Design Science Research (DSR)

Phase 3: Development



Methodology - Design Science Research (DSR)

Phase 4: Quantitative Evaluation

Retrieval accuracy, contextual relevance, response stability

ROUGE: Lexical coverage (Lin, 2004)

BERTScore: Semantic similarity (Zhang et al., 2019)

RAGAS: Faithfulness, context precision, context recall, answer relevance (Es et al., 2024)

Models: Llama2, Llama3 (open-weight) and GPT-5 (proprietary baseline)

Statistical analysis

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Papineni, K. et al. (2002). BLEU: a method for automatic evaluation of machine translation. ACL.

Lin, C-Y. (2004). ROUGE: A package for automatic evaluation of summaries. ACL Workshop.

Zhang, T. et al. (2019). BERTScore: Evaluating text generation with BERT. arXiv:1904.09675.

Es, S. et al. (2024). RAGAS: Automated evaluation of retrieval augmented generation. EAACL.

Methodology - Design Science Research (DSR)

Phase 4: Qualitative Evaluation

User **perception** and **acceptance** through Technology Acceptance Model (TAM).

Participation criteria:

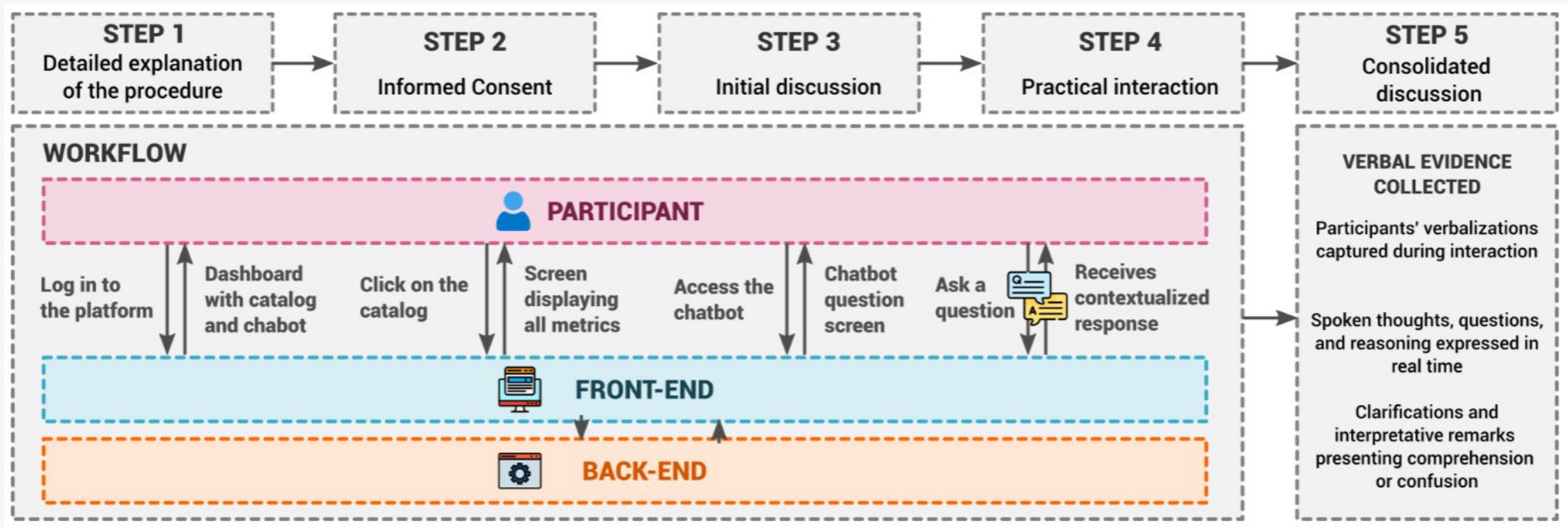
- Aged 18 or older
- Experience in blockchain development and software quality
- At least one year of professional or research experience

Recruitment: LinkedIn + individual email invitations

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Methodology - Design Science Research (DSR)

Phase 4: Qualitative Evaluation



Technology Acceptance Model (TAM) (Davis et al. , 1989): Perceived Usefulness, Perceived Ease of Use, Attitude Toward Use, Behavioral Intention

Methods: Think-Aloud (Rubin & Chisnell, 2011) · Participant Observation (Cooper et al., 2004)

Davis, F. D. et al. (1989). Technology Acceptance Model. Management Science, 35(8).

Rubin, J. & Chisnell, D. (2011). Handbook of Usability Testing. Wiley.

Cooper, J. et al. (2004). Using participant or non-participant observation to explain information behaviour. Information Research.

(i) Methodological approach

- Structured approach combining technical and social evaluation

(ii) Integrated artifact

- Metrics catalog (ISO/IEC 25010) + RAG-based chatbot as a decision-support mechanism

(iii) Evaluation plan

- Replicable basis for assessing blockchain quality with transparency and reproducibility

Next steps:

- Conduct **full quantitative** and **qualitative empirical** studies.
- Incorporate **platform-aware** metric interpretation (e.g., Ethereum vs. Solana).



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